Category: ENGINEERING

ENGINEERING INTERN Skills C++, Python, MATLAB, Git, Bash, R, SQL (basic). Experienced in Linux/Unix and using high performance computing clusters. Machine Learning Tools and Libraries: Scikit-learn, Pandas, Seaborn, matplotlib, TensorFlow (basic). (I built a XGBoost model that has 77.5% accuracy in the Kaggle Titanic challenge.) Computational Fluid Dynamics and Discrete Element Method Codes CFD-DEM, OpenFOAM, CFD-ACE+®, Fluent®, COMSOL®, LAMMPS, and LIGGGHTS. Reservoir and Fracture Modeling Tools CMG® for reservoir simulation; FracPro® for fracture simulation and analysis; Saphir for pressure transient analysis. Experimental and Statistical Methods SEM, AFM, Confocal Microscopy, Regression analysis, Statistical process control, Design of experiments. Experience ENGINEERING INTERN 08/2016 ■ 12/2016 Company Name State Project: Develop a cavings transport model for optimizing hole-cleaning operations. Developed a solids transport model for predicting cuttings/cavings bed height during a hole-cleaning operation. In
contrast to conventional CFD models that typically take several hours to run, this novel numerical model can obtain
results within a few minutes, enabling timely optimization of the well circulation schedule. Investigated the competitive landscape and designed the commercialization plan for the numerical model. Leveraged the experiences from internal drilling experts and aligned with all stakeholders throughout the development
process. ENGINEERING INTERN 05/2016 ■ 08/2016 City , State Project: Optimize diverter pumping schedule for better production performance after well re-stimulation. Built a simulator to model proppant, diverter, and slurry distribution in a plug-and-perf hydraulic fracturing operation. Derived a simple proxy model to substitute time-consuming CFD-DEM simulations for predicting diverter transport
through perforation clusters. Simulation time drops from 48+ hours to less than 1 sec. Provided recommendations for pumping schedule design in a fracturing treatment. PROCESS ENGINEER 04/2012 ■ 05/2013 Company Name City Improve display yield through statistical modeling, process control, and tool modifications. Won Qualstar award in Nov. 2012 by completing two specific yield improvement tasks in merely two months, first time
for QMT-TW to award its engineers after establishment. Optimized sealing process of interferometric modulator (iMoD) display that led to 52% pre-functional yield increase. Increased the up time of panel encapsulation station from 73% to 92% by leading two tool-modification projects involving a group of 5 equipment engineers and 2 external support engineers from Japan. PROPPANT/DIVERTER TRANSPORT in HORIZONTAL WELLS, UT Austin Aug. 2014-present. Evaluate the efficiency of proppant/diverter transport in perforated horizontal wells under different slurry flow
conditions using a combined CFD-DEM approach. Developed a multivariate statistical model to substitute traditional CFD model for predicting proppant transport
through perforations at various flow conditions. The computational cost dropped 5 orders of magnitude. Accurately predicted DAS-measured proppant distribution in a field case with less than 10% error. Chu-Hsiang Wu Page 2
DESIGN and SELECTION of GRAVEL PACK and SAND CONTROL SCREENS, UT Austin Jun. 2013-present. Develop analytical, DEM, and Monte Carlo models for predicting sand production through gravel packs and sand
control screens. Accurately predicted 6 sand production lab-test results obtained from operators with less than 15% error using the
developed model. Invented a DEM-based approach for extracting pore throat size distribution of complex packings. Results show that the
pore throat sizes within gravel packs are usually between 1/5 to 1/9 of the effective gravel size. The findings
correspond remarkably well with previous field observations and enable further optimization of gravel pack designs. 06/2010 Company Name Designed highly mixing-efficient serpentine channels for biomedical detection. Devised mixing index to quantify mixing efficiency of two heterogeneous fluids flowing in microfluidic channels. Interests SPE translator, Nepal medical assistance group, cofounder of Taiwan Bio-Nano Youth Initiative.
SELECTED PUBLICATIONS

· Wu, C.-H., Sharma, M. M. 2017. A DEM-based approach for evaluating the pore throat size distribution of a filter
medium, Powder Technology, ISSN 0032-5910, https://doi.org/10.1016/j.powtec.2017.09.018.
· Wu, C.-H., Sharma, M. M., Chanpura, R. et al. 2017. Factors Governing the Predicted Performance of Multilayered
Metal-Mesh Screens. SPE Drilling & Completion. SPE-178955-PA.
https://doi.org/10.2118/178955-PA.
· Wu, C.-H., Sharma, M. M. 2016. Effect of Perforation Geometry and Orientation on Proppant Placement in Perforation
Clusters in a Horizontal Well. Paper SPE-179117-MS was presented at the SPE Hydraulic Fracturing Technology
Conference, The Woodlands, TX, USA, 9-11 February 2016.
Wu, C.-H., Yi, S., Sharma, M. M. 2017. Proppant Distribution Among Multiple Perforation Clusters in a Horizontal
Wellbore. Paper SPE-184861-MS was presented at the SPE Hydraulic Fracturing Technology Conference, The
Woodlands, TX, USA, 24-26 January 2017.
· Mondal, S., Wu, C.-H., Sharma, M. M. et al. 2016. Characterizing, Designing, and Selecting Metal Mesh Screens for
Standalone-Screen Applications. SPE Drill & Compl 31 (2): 85-94. SPE-170935-PA.
http://dx.doi.org/10.2118/170935-PA.
· Mondal, S., Wu, C.-H., Sharma, M. M. 2016. Coupled CFD-DEM Simulation of Hydrodynamic Bridging at Constrictions.
Int. J. Multiph. Flow, Vol. 84, pp. 245-263, ISSN 0301-9322,
http://dx.doi.org/10.1016/j.ijmultiphaseflow.2016.05.001.
· Zhang, K., Chanpura, R. A., Mondal, S., Wu, C.-H., Sharma, M. M., Ayoub, J. A., & Parlar, M. 2015. Particle Size Education and Training May 2018 Ph.D : UT Austin - PETROLEUM ENGINEERING Scientific Computation City , State PETROLEUM ENGINEERING Scientific Computation Dissertation: Modeling Particulate Flows in Conduits and Porous Media; Supervisor: Mukul M. Sharma 3.9/4.0 Recipient of ConocoPhillips Fellowship (2013), and Jack L. Thurber Memorial Endowed Presidential Scholarship Jun. 2010 M.S : National Taiwan University - MECHANICAL ENGINEERING City , Taiwan MECHANICAL ENGINEERING Design of a mixing-efficient microfluidic device for bio-medical applications 3.9/4.0 Jun. 2008 B.S : National Tsing Hua University - POWER MECHANICAL ENGINEERING City , Taiwan POWER MECHANICAL ENGINEERING Presidential Award and Scholarship (2006, 2007) Skills approach, Bash, basic, C++, competitive, DAS, Design of experiments, functional, lab-test, Linux, Machine Learning, MATLAB, Modeling, novel, optimization, process control, Programming, proxy, Python, Simulation, SQL, Statistical process control, Supervisor, Unix Additional Information LEADERSHIP and VOLUNTEER
· Served as a Second Lieutenant in an artillery company in the Taiwan Army during 2010-2011.
· Qualstar Award, Qualcomm, 2012 and 2013
· Qualcomm Know-how Incentive Award, Qualcomm, 2013
· Technical Editor of SPE Journal, SPE Drilling and Completion, SPE Production and Operations, 2017-present
· Volunteer experience: SPE translator, Nepal medical assistance group, cofounder of Taiwan Bio-Nano Youth Initiative.
SELECTED PUBLICATIONS
· Wu, C.-H., Sharma, M. M. 2017. A DEM-based approach for evaluating the pore throat size distribution of a filter
medium, Powder Technology, ISSN 0032-5910, https://doi.org/10.1016/j.powtec.2017.09.018.
· Wu, C.-H., Sharma, M. M., Chanpura, R. et al. 2017. Factors Governing the Predicted Performance of Multilayered
Metal-Mesh Screens. SPE Drilling & Completion. SPE-178955-PA.
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· Mondal, S., Wu, C.-H., Sharma, M. M. et al. 2016. Characterizing, Designing, and Selecting Metal Mesh Screens for
Standalone-Screen Applications. SPE Drill & Compl 31 (2): 85-94. SPE-170935-PA.
http://dx.doi.org/10.2118/170935-PA.
· Mondal, S., Wu, C.-H., Sharma, M. M. 2016. Coupled CFD-DEM Simulation of Hydrodynamic Bridging at Constrictions.
Int. J. Multiph. Flow, Vol. 84, pp. 245-263, ISSN 0301-9322,
http://dx.doi.org/10.1016/j.ijmultiphaseflow.2016.05.001.
· Zhang, K., Chanpura, R. A., Mondal, S., Wu, C.-H., Sharma, M. M., Ayoub, J. A., & Parlar, M. 2015. Particle Size
Distribution Measurement Techniques and Their Relevance or Irrelevance to Sand Control Design. SPE Drill & Compl 30
(2): 164-174. SPE-168152-PA. http://dx.doi.org/10.2118/168152-PA.